EXPERIMENT 4:

AIM: Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char text[100], key[100];
    int i, j = 0;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter key: ");
    scanf("%s", key);

    for (i = 0; text[i] != '\0'; i++) {
        if (isalpha(text[i])) {
            char base = isupper(text[i]) ? 'A' : 'a';
            text[i] = (text[i] - base + tolower(key[j % strlen(key)]) - 'a') % 26 + base;
            j++;
        }
    }
```

```
    printf("Encrypted text: %s", text);
    return 0;
}
```

OUTPUT:

